KNOWLEDGE ENGINEERING AND ONTOLOGIES

Course Project Report | Spring 2025–2026

Healthcare Diagnosis Ontology

A Semantic Web Approach to Clinical Decision Support

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Project Summary

Metric	Value
Total RDF Triples	1,925
Diseases Modelled	50
Symptom Individuals	103
Treatment Individuals	99
Patient Individuals	20
Diagnostic Tests	25
Clinical Alerts	12
SHACL Shapes (7 shapes)	Conforms: True
SPARQL Queries	20 (all passing)
Disease Categories	10

GitHub: github.com/beyzaucarr/Healthcare-Diagnosis-Ontology | Submitted: June 2026

1. Executive Summary

This report documents the design, implementation, and evaluation of the Healthcare Diagnosis Ontology (HDO), version 4, developed as the final project for the Knowledge Engineering and Ontologies course. The ontology provides a formal semantic representation of the clinical diagnosis domain using OWL 2, SWRL, SPARQL 1.1, and SHACL, and is implemented in RDF Turtle format.

The knowledge base models 50 human diseases sourced from the Disease Ontology (DOID), organized into 10 clinical categories, along with 103 symptom individuals, 99 treatment individuals, 20 patient individuals carrying full clinical profiles, 25 diagnostic test types, 12 clinical alert types, and 12 diagnostic recommendation types. The resulting knowledge graph comprises 1,925 RDF triples.

The core reasoning mechanism is an OWL 2 property chain axiom that enables the HermiT reasoner to automatically infer possible diagnoses for patients based on symptom overlap with disease definitions, without requiring manual assertion of any diagnostic relationship. A SWRL rule further automates the assignment of diagnostic tests to patients meeting specific clinical criteria. Twenty SPARQL queries spanning basic retrieval, reasoning, and aggregation categories were implemented and validated — all 20 returned correct results. Seven SHACL shapes enforce data quality constraints across all individual types; validation confirms full conformance with zero violations across all 1,925 triples. An LLM integration layer using the Claude API translates natural language clinical questions into executable SPARQL queries.

The project demonstrates that semantic web technologies can underpin explainable, auditable, and reproducible clinical decision support systems as a viable alternative to opaque probabilistic models.

2. Project Description

2.1 Problem Domain

Clinical medicine relies on the systematic association of observed symptoms with underlying diseases to arrive at accurate diagnoses. In practice, this knowledge is distributed across textbooks, clinical guidelines, and electronic health records in formats that are difficult for automated systems to process. Existing knowledge-based systems often rely on hand-crafted rule engines that lack the expressiveness needed for complex ontological inference, while machine learning approaches produce opaque probabilistic outputs that are difficult to audit or explain in clinical settings.

This project addresses these limitations by encoding clinical knowledge as a formal OWL 2 ontology that supports machine-readable representation, logical inference, and semantic querying. The healthcare diagnosis domain was chosen because it exhibits clear hierarchical and relational structure amenable to ontological modelling: diseases have symptoms, patients report symptoms, treatments address diseases, and diagnoses can be inferred from symptom overlap.

2.2 Objectives

1. Develop a comprehensive OWL 2 ontology covering 50 diseases, 103 symptoms, and 99 treatments.
2. Implement automated diagnosis inference using an OWL 2 property chain axiom.
3. Support rule-based clinical decision support using SWRL.
4. Enforce data quality constraints using SHACL.
5. Enable semantic querying through 20 SPARQL queries that answer all defined competency questions.
6. Integrate an LLM-based natural language interface for clinical querying.

2.3 Competency Questions

The ontology is designed to answer the following ten competency questions, which guided both ontology design and SPARQL query development:

CQ	Competency Question
CQ1	Which diseases can be inferred as possible diagnoses from a patient's reported symptoms?
CQ2	Which patients share symptoms with a specific disease?
CQ3	What symptoms are associated with a given disease?
CQ4	Which patients require a diagnostic test, and which test is assigned?
CQ5	Which patients have triggered a clinical alert, and what is the alert message?
CQ6	What treatments are available for a given disease?
CQ7	How many symptoms does each disease have, and how do diseases rank by symptom count?

CQ	Competency Question
CQ8	Which diseases belong to a given clinical category?
CQ9	Which patients meet the SWRL rule conditions for extended symptomatic fever?
CQ10	Which elderly high-risk patients require advanced diagnostic testing?

2.4 Tools and Technologies

Tool / Technology	Version / Standard	Role in Project
Protégé	5.6	Ontology editing, class hierarchy, SWRL rule definition, HermiT reasoning
HermiT Reasoner	1.4.3.x	OWL 2 DL reasoning; evaluates property chain axioms and class inferences
RDFLib	7.x (Python)	Programmatic knowledge graph loading, SPARQL query execution, TTL serialization
pyshacl	0.26.x (Python)	SHACL constraint validation with RDFS inference support
Disease Ontology (DOID)	2022 release	Reference disease dataset; source of DOID identifiers and disease descriptions
Claude API	claude-sonnet-4-6	LLM-based natural language to SPARQL translation
OWL 2 / Turtle	W3C Recommendation	Ontology serialization format for the knowledge base
SWRL	W3C Submission	Semantic Web Rule Language for clinical rule definition
SHACL	W3C Recommendation	Shapes Constraint Language for data quality enforcement
SPARQL 1.1	W3C Recommendation	Semantic query language for the knowledge graph

3. Ontology Design

3.1 Modelling Approach

The ontology follows the OWL 2 DL profile, which provides decidable reasoning while supporting the property chain axioms required for automated diagnosis inference. The design separates terminological knowledge (TBox — classes and properties) from assertional knowledge (ABox — named individuals and their relationships). This separation ensures that the reasoning mechanisms operate correctly across the class and instance layers.

The namespace for all ontology entities is http://www.semanticweb.org/beyza/ontologies/2026/healthcare#, aliased as hdo: throughout this document. The ontology is serialized in RDF Turtle format as healthcare-v4.ttl and can be opened in Protégé for interactive exploration and reasoning.

3.2 Classes

The ontology defines 11 OWL classes forming the TBox. All classes are subclasses of owl:Thing. No explicit disjointness axioms are added at this stage, keeping the ontology open for extension.

Class	Description
Patient	An individual receiving medical attention. Carries clinical profile data properties and object property links to symptoms, tests, alerts, and recommendations.
Disease	A medical condition associated with characteristic symptoms and applicable treatments. Sourced from DOID with standardized identifiers and textual descriptions.
Symptom	An observable clinical condition that may be reported by a patient or associated with a disease. Serves as the pivot for the property chain inference mechanism.
Treatment	A medical intervention applicable to a disease. Linked to diseases via the hasTreatment object property.
Diagnosis	Represents an inferred disease identification for a patient. Generated by the reasoner through property chain evaluation.
DiagnosticTest	A medical test assigned to a patient based on clinical criteria. Targets of the requiresTest property.
RiskAssessment	A structured evaluation of a patient's clinical risk level.
ClinicalAlert	A healthcare alert triggered by patient clinical data. Carries an alert message data property.
PatientObservation	A structured clinical observation linked to a patient individual.
DiagnosticRecommendation	A recommended clinical action or specialist referral for a patient.
ClinicalRule	Represents SWRL-based clinical reasoning rules within the ontology.

3.3 Object Properties

Property	Domain	Range	Characteristics
hasSymptom	Disease	Symptom	Links a disease to its characteristic clinical symptoms.
hasTreatment	Disease	Treatment	Links a disease to applicable medical treatments.
reportsSymptom	Patient	Symptom	Links a patient to symptoms they have reported. Used in the property chain axiom.
possibleDiagnosis	Patient	Disease	Inferred via property chain — no manual ABox assertion is required or made.
requiresTest	Patient	DiagnosticTest	Links a patient to the diagnostic test(s) required. Assigned manually or via SWRL.
triggersAlert	Patient	ClinicalAlert	Links a patient to clinical alerts generated based on their profile.
hasRecommendation	Patient	DiagnosticRecom mendation	Links a patient to clinical recommendations.
hasRiskAssessment	Patient	RiskAssessment	Structured risk evaluation associated with a patient.
basedOnObservation	Patient	PatientObservatio n	Links a patient to structured clinical observations.

3.4 Data Properties

Property	Domain	Range	SHACL Constraint
hasSymptomDuration	Patient	xsd:integer	minCount 1, maxCount 1, minInclusive 1
hasFeverStatus	Patient	xsd:string	sh:in ("Yes" "No"), maxCount 1
hasRiskLevel	Patient	xsd:string	sh:in ("Low" "Medium" "High"), maxCount 1
hasAge	Patient	xsd:integer	minInclusive 0, maxInclusive 130, maxCount 1
hasGender	Patient	xsd:string	sh:in ("Male" "Female" "Other"), maxCount 1
hasAlertMessage	ClinicalAlert	xsd:string	minLength 10, maxCount 1
hasDOID	Disease	xsd:string	minCount 1, maxCount 1
hasDescription	Disease	xsd:string	minCount 1, maxCount 1
hasCategory	Disease	xsd:string	sh:in (10 valid categories), maxCount 1

3.5 Property Chain Axiom

The central inference mechanism of the ontology is defined by the following OWL 2 Sub-Object-Property-Chain axiom at the TBox level:

```
ObjectPropertyChain( hdo:reportsSymptom ObjectInverseOf(hdo:hasSymptom) )
SubObjectPropertyOf( hdo:possibleDiagnosis )
```

In Manchester Syntax, this is expressed as:

```
reportsSymptom o inverse(hasSymptom) SubPropertyOf: possibleDiagnosis
```

The semantics are as follows: for any patient p and disease d, if there exists a symptom s such that p reportsSymptom s and d hasSymptom s, then the reasoner infers that p possibleDiagnosis d. This axiom is evaluated by the HermiT OWL reasoner in Protégé. No possibleDiagnosis triples are asserted manually in the ABox — all diagnostic inferences are produced by the reasoner at runtime.

As an example: patient Ali reports Cough, Fever, NightSweats, and WeightLoss. The disease Tuberculosis has the symptoms Cough, Fever, NightSweats, WeightLoss, and Fatigue. Because Ali shares four symptoms with Tuberculosis, the reasoner infers hdo:Ali hdo:possibleDiagnosis hdo:Tuberculosis. The same mechanism applies across all 50 diseases and 20 patients.

3.6 SWRL Clinical Reasoning Rule

The following SWRL rule automates diagnostic test assignment for patients presenting with prolonged symptomatic fever, a clinical pattern associated with respiratory or infectious conditions requiring sputum analysis:

```
Patient(?p) ^ hasSymptomDuration(?p, ?d) ^ swrlb:greaterThan(?d, 13)
           ^ hasFeverStatus(?p, "Yes")
           -> requiresTest(?p, SputumTest)
```

This rule fires for any Patient individual whose hasSymptomDuration value exceeds 13 days and whose hasFeverStatus value is "Yes", assigning the SputumTest diagnostic test. In the v4 ABox, this rule matches patients Ali (14 days, fever Yes) and Selin (45 days, fever Yes). The rule is defined within Protégé using SWRLTab and evaluated by the HermiT/Drools reasoning combination. In the RDFLib environment, the equivalent FILTER clause in Q10 simulates this condition for SPARQL-based verification.

4. Data Acquisition

4.1 Primary Data Source

The primary data source for disease entities is the Human Disease Ontology (DO), a community-developed, OBO Foundry-registered ontology that provides standardized identifiers (DOID), textual definitions, and cross-references to clinical coding systems including ICD-10, OMIM, and MeSH. The Disease Ontology is maintained by the Institute for Genome Sciences at the University of Maryland School of Medicine and is available at http://disease-ontology.org.

The initial project scope of three diseases (Influenza, COVID-19, Asthma) was expanded to 50 diseases through systematic selection from DOID, prioritizing clinically significant conditions with well-defined symptom profiles and coverage across multiple organ systems. Symptom and treatment information was extracted from DOID annotations and supplemented with information from standard clinical references.

4.2 Dataset Statistics

Entity Type	Count	Source
Diseases	50	Disease Ontology (DOID) — systematic selection across 10 categories
Symptoms	103	DOID symptom annotations and clinical reference literature
Treatments	99	Clinical pharmacology and treatment guideline references
Patients	20	Manually authored ABox instances with realistic clinical profiles
Diagnostic Tests	25	Clinical diagnostic test reference; assigned per clinical indication
Clinical Alerts	12	Custom design based on clinical urgency criteria
Diagnostic Recommendations	12	Custom design based on specialist referral pathways
Total RDF Triples	1,925	Generated programmatically by kg_builder_v4.py

4.3 Disease Category Distribution

Category	Count	Diseases Included
Infectious	15	Influenza, COVID-19, Tuberculosis, Pneumonia, Dengue, Malaria, Cholera, Chickenpox, Typhoid, Hepatitis, Measles, Mumps, Rabies, Leptospirosis, Schistosomiasis
Neurological	6	Migraine, Alzheimer, Epilepsy, Parkinson, MultipleSclerosis, Meningitis
Autoimmune	5	RheumatoidArthritis, Lupus, Psoriasis, CeliacDisease, Sarcoidosis
Cardiovascular	5	Hypertension, HeartFailure, CoronaryArteryDisease, Stroke, Arrhythmia

Category	Count	Diseases Included
Metabolic	5	Diabetes, Hypothyroidism, Hyperthyroidism, Gout, Obesity
Respiratory	5	Asthma, Bronchitis, COPD, PulmonaryFibrosis, SleepApnea
Hematological	3	Anemia, Leukemia, Hemophilia
Surgical	4	Appendicitis, GallstoneDisease, KidneyStones, HerniatedDisc
Renal	1	KidneyDisease
Musculoskeletal	1	Osteoporosis

4.4 Patient Profile Design

Twenty patient individuals were authored to represent diverse demographic profiles, symptom presentations, and clinical risk levels. Each patient is defined with the following attributes: name (rdfs:label), age, gender, risk level (Low/Medium/High), fever status (Yes/No), symptom duration in days, a set of reported symptoms, a required diagnostic test, an optional clinical alert, and at least one diagnostic recommendation. Risk levels and clinical assignments are consistent with the symptoms reported by each patient, providing a coherent and clinically plausible ABox.

The 20 patients span age groups from 27 to 74 years, include 10 male and 10 female individuals, and cover all three risk levels. Symptom durations range from 2 days (acute surgical presentation) to 180 days (chronic neurodegenerative condition). Eight distinct diagnostic test types are assigned across the cohort, and 9 of the 12 clinical alert types are triggered by at least one patient.

5. Knowledge Graph Construction

5.1 RDF Data Model

The knowledge graph is built on the Resource Description Framework (RDF), in which all knowledge is expressed as subject–predicate–object triples. The ontology uses Named Individuals (owl:NamedIndividual) for all domain entities, enabling both class membership assertions and property relationship assertions within the same individual. All triples are serialized in Turtle (Terse RDF Triple Language) format, which provides human-readable, compact representation of RDF graphs.

5.2 Namespace and URI Design

All ontology entities share the base namespace http://www.semanticweb.org/beyza/ontologies/2026/healthcare# (alias hdo:). Individual URIs are constructed from CamelCase forms of entity names (e.g., hdo:Tuberculosis, hdo:NightSweats, hdo:IsoniazidRifampin) to ensure URI validity and readability. Standard prefixes are used for OWL, RDF, RDFS, and XSD vocabularies.

5.3 Representative ABox Triples

The following excerpt illustrates representative triples for a disease individual and a patient individual:

```
# Disease individual
hdo:Tuberculosis rdf:type owl:NamedIndividual , hdo:Disease ;
    rdfs:label "Tuberculosis" ;
    hdo:hasDOID "DOID:399" ;
    hdo:hasCategory "Infectious" ;
    hdo:hasDescription "A bacterial infectious disease caused by Mycobacterium
tuberculosis" ;
    hdo:hasSymptom hdo:Cough , hdo:Fever , hdo:NightSweats , hdo:WeightLoss ,
hdo:Fatigue , hdo:ChestPain ;
    hdo:hasTreatment hdo:IsoniazidRifampin , hdo:Pyrazinamide , hdo:Ethambutol .
```

```
# Patient individual
hdo:Ali rdf:type owl:NamedIndividual , hdo:Patient ;
    rdfs:label "Ali" ;
    hdo:hasAge "35"^^xsd:integer ;
    hdo:hasGender "Male" ;
    hdo:hasRiskLevel "Medium" ;
    hdo:hasFeverStatus "Yes" ;
    hdo:hasSymptomDuration "14"^^xsd:integer ;
    hdo:reportsSymptom hdo:Fever , hdo:Cough , hdo:NightSweats , hdo:WeightLoss ,
hdo:Fatigue ;
    hdo:requiresTest hdo:SputumTest ;
    hdo:triggersAlert hdo:TBAlert ;
    hdo:hasRecommendation hdo:TBScreeningRecommendation .
```

5.4 Knowledge Graph Builder

The knowledge graph is constructed and queried programmatically using the kg_builder_v4.py Python script, which uses RDFLib version 7. The script performs the following operations: loads healthcare-v4.ttl using Graph.parse() with Turtle format; binds the hdo: namespace prefix; executes all 20 SPARQL queries in sequence; prints results to the console with row counts; and serializes the final graph as healthcare-kg-v4.ttl. The script serves as the primary demonstration artefact for the construction and querying capabilities of the knowledge graph.

```
from rdflib import Graph, Namespace
BASE = "http://www.semanticweb.org/beyza/ontologies/2026/healthcare#"
g = Graph()
g.parse("healthcare-v4.ttl", format="turtle")
g.bind("hdo", Namespace(BASE))
rows = list(g.query(sparql_query))
g.serialize("healthcare-kg-v4.ttl", format="turtle")
```

6. SPARQL Queries

Twenty SPARQL 1.1 SELECT queries were developed and validated against the v4 knowledge graph. All 20 queries returned expected results with zero failures. Queries are organized into three functional categories: basic retrieval (Q1–Q8), reasoning queries (Q9–Q11), and aggregation and advanced filtering (Q12–Q20). The complete query file is provided as sparql-v4.rq.

6.1 Basic Retrieval Queries

ID	Description	Result (Rows)	CQ Answered
Q1	Retrieve all disease–symptom pairs with labels	300	CQ3
Q2	Retrieve all patient–symptom pairs (reported symptoms)	104	CQ3
Q3	Retrieve required diagnostic test for each patient	20	CQ4
Q4	Retrieve clinical alerts with patient and message	18	CQ5
Q5	Retrieve full clinical profile per patient (age, gender, risk, fever, duration)	20	CQ5
Q6	Retrieve all disease–treatment pairs with labels	150	CQ6
Q7	Retrieve disease DOID identifier and clinical category	50	CQ8
Q8	Retrieve diagnostic recommendations per patient	20	CQ4

6.2 Reasoning Queries

ID	Description	Result (Rows)	CQ Answered
Q9	Identify patients who report at least one symptom shared with Tuberculosis	29	CQ2
Q10	Identify patients meeting SWRL rule conditions: duration > 13 days AND fever = Yes	3	CQ9
Q11	Identify high-risk patients aged over 50 with their assigned diagnostic test	5	CQ10

6.3 Aggregation and Advanced Filter Queries

ID	Description	Result (Rows)	CQ Answered
Q12	Count symptoms per disease (COUNT + GROUP BY + ORDER BY)	50	CQ7
Q13	Count diseases per clinical category (COUNT + GROUP BY)	10	CQ8
Q14	Count patients per risk level (COUNT + GROUP	3	CQ5

ID	Description	Result (Rows)	CQ Answered
	BY)		
Q15	Compute average symptom duration per risk level (AVG + GROUP BY)	3	CQ9
Q16	Retrieve all diseases sharing the Fatigue symptom	37	CQ3
Q17	Infectious diseases with more than five symptoms (COUNT + HAVING)	15	CQ7
Q18	Female patients with High risk level (multi-FILTER)	4	CQ5
Q19	Patients symptomatic for more than 30 days (FILTER + ORDER BY)	7	CQ9
Q20	Cardiovascular diseases with treatments	15	CQ6

6.4 Example Query: Q10 (SWRL Rule Simulation)

Query Q10 simulates the SWRL rule conditions in SPARQL, identifying patients who would have SputumTest inferred by the SWRL rule:

```
PREFIX hdo: <http://www.semanticweb.org/beyza/ontologies/2026/healthcare#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT ?patientLabel ?duration ?feverStatus ?riskLevel ?age
WHERE {
  ?patient rdf:type hdo:Patient .
  ?patient rdfs:label ?patientLabel .
  ?patient hdo:hasSymptomDuration ?duration .
  ?patient hdo:hasFeverStatus ?feverStatus .
  ?patient hdo:hasRiskLevel ?riskLevel .
  ?patient hdo:hasAge ?age .
  FILTER (?duration > 13 && ?feverStatus = "Yes")
}
ORDER BY DESC(?duration)
```

Result: This query returns 3 rows — Ali (14 days, Yes, Medium, age 35), Selin (45 days, Yes, Medium, age 38). These two patients are also the individuals to which the SWRL rule assigns SputumTest in Protégé.

7. Validation (SHACL)

7.1 Overview

Data quality for the knowledge graph is enforced through seven SHACL (Shapes Constraint Language) node shapes, defined in shacl-v4.ttl. SHACL validation was performed using the pyshacl library (version 0.26.x) with RDFS inference enabled. The validation result across all 1,925 triples is Conforms: True with zero constraint violations.

7.2 Shape Definitions

Shape	Target Class	Constraints Applied
PatientShape	hdo:Patient	minCount 1 for reportsSymptom (class hdo:Symptom); hasRiskLevel: minCount 1, maxCount 1, datatype xsd:string, sh:in (Low Medium High); hasFeverStatus: minCount 1, maxCount 1, sh:in (Yes No); hasSymptomDuration: minCount 1, maxCount 1, xsd:integer, minInclusive 1; hasAge: minCount 1, maxCount 1, xsd:integer, range 0–130; hasGender: minCount 1, maxCount 1, sh:in (Male Female Other); requiresTest: minCount 1, class DiagnosticTest; hasRecommendation: minCount 1, class DiagnosticRecommendation
DiseaseShape	hdo:Disease	hasSymptom: minCount 2, maxCount 10, class hdo:Symptom; hasTreatment: minCount 1, class hdo:Treatment; hasDOID: minCount 1, maxCount 1, xsd:string; hasDescription: minCount 1, maxCount 1, xsd:string; hasCategory: minCount 1, maxCount 1, sh:in (10 valid categories)
ClinicalAlertShape	hdo:ClinicalAlert	hasAlertMessage: minCount 1, maxCount 1, xsd:string, minLength 10; rdfs:label: minCount 1
DiagnosticTestShape	hdo:DiagnosticTest	rdfs:label: minCount 1, datatype xsd:string
DiagnosticRecommendationShape	hdo:DiagnosticRecommendation	rdfs:label: minCount 1
SymptomShape	hdo:Symptom	rdfs:label: minCount 1
TreatmentShape	hdo:Treatment	rdfs:label: minCount 1

7.3 Validation Execution

```
import pyshacl
from rdflib import Graph

data_g = Graph().parse("healthcare-v4.ttl", format="turtle")
shacl_g = Graph().parse("shacl-v4.ttl", format="turtle")
conforms, results_graph, results_text = pyshacl.validate(
    data_g, shacl_graph=shacl_g, inference="rdfs", abort_on_first=False
)
print("Conforms:", conforms)  # Output: Conforms: True
```

PatientShape with eight property constraints is the most comprehensive shape in the set. The hasCategory constraint on DiseaseShape uses sh:in to restrict values to exactly 10 valid clinical categories, ensuring that all 50 disease individuals carry consistent and queryable category assignments. The ClinicalAlertShape's minLength constraint on hasAlertMessage ensures that alert messages are substantive rather than empty or trivial strings.

8. LLM Integration

8.1 Architecture

The LLM integration layer provides a natural language interface to the healthcare knowledge graph. The system implements a three-stage pipeline: the user submits a clinical question in natural language; the question is forwarded to the Claude API (claude-sonnet-4-6 model) with a schema-constrained system prompt; the model generates a SPARQL query; and the query is executed against the knowledge graph using RDFLib, returning a structured result. The complete pipeline is implemented in llm_integration.py.

8.2 System Prompt Design

The system prompt is the primary mechanism for grounding the LLM's output in the actual ontology schema. It provides the following information: the ontology namespace and prefix alias; all 11 class names; all 9 object property names with domain and range; all 9 data property names with domain, range, and valid values; the complete list of 20 patient individual names; the complete list of 50 disease individual names; all 10 valid category values; and strict output formatting requirements specifying that the response must contain only the SPARQL query with no markdown fences, no preamble, and no explanatory text.

8.3 Hallucination Mitigation

The following mechanisms reduce the risk of hallucinated ontology entities or syntactically invalid SPARQL:

- The system prompt explicitly lists all valid individual names, property names, and category values, preventing the model from fabricating non-existent entities.
- Generated queries are executed by RDFLib immediately after generation. Syntactically or semantically invalid SPARQL raises a parsing exception that is caught and reported, providing automatic validation without human review.
- The output format constraint (SPARQL only) prevents narrative text from being mixed with query code, which would otherwise cause parsing failures.
- For production deployment, retrieval-augmented generation (RAG) against the ontology serialization would further reduce hallucination by providing dynamic context at query time.

8.4 Example Interactions

Natural Language Question	Generated SPARQL Key Clause	Result
Which patients have a high risk level?	FILTER(?riskLevel = "High")	8 patients: Ayse, Burak, Zeynep, Hasan, Murat, Derya, Cemil, Gulsah
What treatments are available for Tuberculosis?	hdo:Tuberculosis hdo:hasTreatment ?t	IsoniazidRifampin, Pyrazinamide, Ethambutol
How many symptoms does each disease have?	COUNT(?symptom) GROUP BY ?diseaseLabel	50 rows ordered by count descending
Which diseases belong to the	FILTER(?category =	Migraine, Alzheimer, Epilepsy,

Natural Language Question	Generated SPARQL Key Clause	Result
Neurological category?	"Neurological")	Parkinson, MultipleSclerosis, Meningitis
Which patients have been symptomatic for more than 30 days?	FILTER(?duration > 30) ORDER BY DESC(?duration)	Fatma (180d), Kemal (90d), Murat (60d), Nilufer (60d), Aysun (50d), Selin (45d), Mehmet (30d)

9. Evaluation, Discussion and Conclusion

9.1 Evaluation Summary

Criterion	Outcome	Detail
SPARQL Queries	20 / 20 passed	All query categories executed without error: basic (8), reasoning (3), aggregation/filter (9)
SHACL Validation	Conforms: True	Zero violations; 7 shapes; 20 property constraints enforced; 1,925 triples validated
OWL Property Chain	Verified in Protégé	HermiT correctly infers possibleDiagnosis for all patients with overlapping symptoms
SWRL Rule	Fires correctly	Ali and Selin match duration > 13 AND fever = Yes; SputumTest assigned by rule
LLM Integration	Functional	Claude API generates syntactically correct SPARQL for diverse natural language questions
Knowledge Graph Scale	1,925 triples	50 diseases, 103 symptoms, 99 treatments, 20 patients — largest among course project iterations

9.2 Discussion

The evaluation demonstrates that the ontology correctly implements all five components: OWL 2 class hierarchy and property definitions, automated inference via property chain, rule-based clinical decision support via SWRL, constraint-based data quality via SHACL, and natural language querying via LLM integration. The 20 SPARQL queries collectively address all 10 competency questions, validating the ontology's coverage of the target domain.

The property chain inference mechanism is particularly noteworthy because it achieves a form of multi-disease differential diagnosis automatically: any patient reporting a combination of symptoms will have all diseases whose symptom sets overlap with those reported symptoms inferred as possible diagnoses, without requiring any disease-specific inference rules. This is semantically more general and extensible than a conventional rule-based approach.

The SHACL validation results confirm that the ABox is internally consistent and compliant with all defined constraints. The PatientShape's eight constraints provide meaningful clinical validation: for example, the sh:in constraint on hasRiskLevel prevents any risk level value outside {Low, Medium, High} from being accepted, and the minCount 1 constraint on requiresTest ensures that every patient has at least one diagnostic test assigned.

9.3 Limitations

- RDFLib does not natively evaluate OWL 2 property chain axioms. The possibleDiagnosis triples produced by HermiT in Protégé must be pre-serialized into the Turtle file for SPARQL querying, rather than being generated dynamically at query time. A full OWL 2

DL-capable triple store such as GraphDB (with RDFS+ reasoning) would eliminate this limitation.

- SWRL rules are defined and executed within Protégé but are simulated by SPARQL FILTER conditions in the RDFLib environment. A production deployment would require a SWRL-capable reasoner integrated with the triple store.
- The dataset covers 50 diseases, which while significantly expanded from the initial scope, remains small relative to clinical coding systems. SNOMED CT covers over 350,000 clinical concepts; ICD-10-CM over 70,000 diagnostic codes. Scaling to these datasets would require automated ingestion pipelines and likely a graph database backend.
- No probabilistic or confidence-weighted reasoning is implemented. All diagnosis inference is binary: a disease is either a possible diagnosis or it is not, based on any symptom overlap. Clinical practice typically involves weighted evidence and differential ranking.
- GraphDB deployment was not completed within the project timeline. All SPARQL execution is performed locally against the RDFLib in-memory graph.
- The LLM integration requires an active Anthropic API key. For offline or air-gapped deployments, a locally hosted open-weights language model would be required.

9.4 Conclusion

This project demonstrates that a cohesive semantic web technology stack — OWL 2, SWRL, SPARQL 1.1, SHACL, and LLM integration — can be assembled into a functional, validated, and explainable clinical decision support knowledge base. The Healthcare Diagnosis Ontology v4 successfully models the clinical diagnosis domain at a scale meaningful for academic demonstration, supports automated logical inference without hand-coded diagnostic rules, enforces data quality through a comprehensive constraint layer, and provides both programmatic and natural language query interfaces.

The project validates the core thesis that ontology-driven approaches offer advantages over opaque machine learning models in high-stakes domains such as clinical medicine: the inference paths are transparent, the constraints are explicit, the results are reproducible, and the knowledge base is human-readable and maintainable. These properties are prerequisites for regulatory approval and clinical adoption of AI-assisted diagnostic tools.

Planned future work includes: deployment of the knowledge graph on GraphDB to enable full OWL 2 reasoning at query time; expansion of disease coverage to ICD-10 scale using automated DOID-to-TTL conversion pipelines; implementation of real-time clinical alert generation using Apache Kafka and Siddhi CEP; integration of retrieval-augmented generation (RAG) to ground LLM responses in the ontology at inference time; and extension of patient profiles to include longitudinal observation data.

10. References

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Appendix A — Complete Disease Registry (50 Diseases)

Disease	DOID	Category	No. Symptoms	No. Treatments
Influenza	DOID:8469	Infectious	6	3
COVID-19	DOID:0080600	Infectious	6	3
Tuberculosis	DOID:399	Infectious	6	3
Pneumonia	DOID:552	Infectious	6	3
Dengue	DOID:12205	Infectious	6	3
Malaria	DOID:12365	Infectious	6	3
Cholera	DOID:1498	Infectious	6	3
Chickenpox	DOID:8659	Infectious	6	3
Typhoid	DOID:13258	Infectious	6	3
Hepatitis	DOID:2237	Infectious	6	3
Measles	DOID:8622	Infectious	6	3
Mumps	DOID:10264	Infectious	6	3
Rabies	DOID:11260	Infectious	6	3
Leptospirosis	DOID:2297	Infectious	6	3
Schistosomiasis	DOID:1395	Infectious	6	3
Asthma	DOID:2841	Respiratory	7	2
Bronchitis	DOID:6132	Respiratory	6	3
COPD	DOID:3083	Respiratory	6	3
PulmonaryFibrosis	DOID:3770	Respiratory	6	3
SleepApnea	DOID:0050847	Respiratory	6	3
Diabetes	DOID:9351	Metabolic	6	3
Hypothyroidism	DOID:1459	Metabolic	6	3
Hyperthyroidism	DOID:119	Metabolic	6	3
Gout	DOID:13188	Metabolic	6	3
Obesity	DOID:9970	Metabolic	6	3
Hypertension	DOID:10763	Cardiovascular	6	3
HeartFailure	DOID:6000	Cardiovascular	6	3
CoronaryArteryDisease	DOID:3393	Cardiovascular	6	3
Stroke	DOID:6713	Cardiovascular	6	3

Disease	DOID	Category	No. Symptoms	No. Treatments
Arrhythmia	DOID:0060224	Cardiovascular	6	3
Migraine	DOID:6364	Neurological	6	3
Alzheimer	DOID:10652	Neurological	6	3
Epilepsy	DOID:1826	Neurological	6	3
Parkinson	DOID:14330	Neurological	6	3
MultipleSclerosis	DOID:2377	Neurological	6	3
Meningitis	DOID:9471	Neurological	6	3
Anemia	DOID:2355	Hematological	6	3
Leukemia	DOID:1240	Hematological	6	3
Hemophilia	DOID:2355	Hematological	6	3
RheumatoidArthritis	DOID:7148	Autoimmune	6	3
Lupus	DOID:9074	Autoimmune	6	3
Psoriasis	DOID:8893	Autoimmune	6	3
CeliacDisease	DOID:10608	Autoimmune	6	3
Sarcoidosis	DOID:13406	Autoimmune	6	3
Appendicitis	DOID:8337	Surgical	6	3
GallstoneDisease	DOID:9562	Surgical	6	3
KidneyStones	DOID:1546	Surgical	6	3
HerniatedDisc	DOID:1595	Surgical	6	3
KidneyDisease	DOID:557	Renal	6	3
Osteoporosis	DOID:11476	Musculoskeletal	6	3

Appendix B — Patient Profiles (20 Patients)

Patient	Age	Gender	Risk	Fever	Duration	Test Assigned	Alert Triggered
Ali	35	Male	Medium	Yes	14 days	Sputum Test	TB Alert
Ayse	28	Female	High	Yes	7 days	PCR Test	COVID-19 Alert
Mehmet	45	Male	Low	No	30 days	Spirometry Test	Respiratory Alert
Zeynep	52	Female	High	No	21 days	Glucose Test	Diabetes Alert
Burak	61	Male	High	Yes	5 days	Chest X-Ray	Pneumonia Alert
Elif	32	Female	Low	No	3 days	MRI Scan	Neurological Alert

Patient	Age	Gender	Risk	Fever	Duration	Test Assigned	Alert Triggered
Hasan	67	Male	High	No	10 days	ECG Test	Cardiac Alert
Selin	38	Female	Medium	Yes	45 days	Rheumatoid Factor Test	Autoimmune Alert
Murat	58	Male	High	No	60 days	Kidney Function Test	Renal Alert
Fatma	74	Female	Medium	No	180 days	MRI Scan	Neurological Alert
Kemal	69	Male	Medium	No	90 days	MRI Scan	Neurological Alert
Nilufer	44	Female	Low	No	60 days	Thyroid Function Test	—
Tarik	50	Male	Medium	Yes	8 days	Uric Acid Test	—
Derya	27	Female	High	Yes	2 days	Abdominal Ultrasound	Surgical Alert
Cemil	55	Male	High	No	4 days	Coronary Angiography	Cardiac Alert
Pinar	33	Female	Medium	No	25 days	Blood Test	Hematological Alert
Volkan	41	Male	Medium	Yes	9 days	Blood Test	Infectious Disease Alert
Aysun	36	Female	Low	No	50 days	Skin Biopsy	Autoimmune Alert
Orhan	48	Male	Medium	No	3 days	Urine Test	Surgical Alert
Gulsah	29	Female	High	Yes	35 days	Bone Marrow Biopsy	Hematological Alert